

Cost-aware graph augmentation for robust networks via effective resistance

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Abstract—Enhancing the robustness of deployed communication and transportation networks against failures and disruptions is critical for reliable operation. We quantify total effective resistance as a tractable robustness proxy and seek to reduce it by adding new links to the network. However, in many applications, this leads to a computationally challenging joint design problem that couples discrete edge selection with continuous weight allocation, while accounting for heterogeneous per-unit link costs. We address this selection-and-sizing combinatorial problem for augmenting a given weighted base graph, and obtain a semidefinite relaxation that yields a convex certificate-producing surrogate. The relaxation is recast as a standard cone program and solved efficiently using a first-order operator-splitting method. A rounding procedure over the relaxed solution yields a feasible discrete design with an a posteriori optimality gap. As a scalable baseline, we also propose a greedy algorithm that exploits rank-one updates to accelerate candidate evaluation. Experiments on synthetic and real networks demonstrate that the method produces cost-aware augmentations that significantly reduce effective resistance under a global budget constraint.

Index Terms—Network robustness, total effective resistance, semidefinite programming, operator splitting, Sherman-Morrison rank-one update.

I. INTRODUCTION

Network robustness is central to the design and operation of transportation, communication, and brain networks [1], [2]. It refers to the network’s ability to sustain functionality under disruptions and has been widely studied; see e.g., [3], [4]. A principled way to quantify it is through spectral measures of the graph Laplacian, the algebraic representation of network connectivity. Among such spectral measures, total effective resistance (*Kirchhoff index*) is widely adopted [4]–[6]. Defined as the sum of the reciprocals of the nonzero Laplacian eigenvalues [5], its evaluation and optimization rely on Laplacian pseudoinverses, which become a computational bottleneck on large graphs [7] and motivate scalable formulations.

Efforts to optimize total effective resistance span several lines of research, varying in design scope and algorithmic strategy. For a fixed topology, [5] formulates optimal weight allocation as a semidefinite program (SDP), but does not address edge augmentation. Finding the optimal set of edges

to add is NP-hard [8]; greedy strategies based on rank-one updates of the Laplacian pseudoinverse are scalable but lack optimality guarantees [9]. Recent work provides greedy approximation guarantees via submodularity-ratio arguments in unweighted settings [10]; however, link addition is not submodular in general and this ratio can be arbitrarily small [11], weakening such guarantees. In learning contexts, [12] rewires graphs to improve graph neural network (GNN) connectivity, and [13] provides theory linking effective-resistance reductions to improved GNN performance. On the application side, [1] studies route addition in air-transportation networks (without weight sizing), and [2] proposes robust inference of brain-network topology. Most closely related to our setting, [14] jointly optimizes link selection and weights under a budget via an SDP relaxation, but designs the network from scratch.

Building on this line, we focus on the *augmentation* regime of an already deployed weighted base network. In this framework the existing edges and their weights are *immutable*, and the design variables are restricted to selecting and sizing a small number of *new* links from a candidate set. Upgrade decisions must be evaluated by their *marginal impact on the already-operational network*, under heterogeneous per-unit costs, explicit weight bounds, and an exactly- k cardinality constraint. Consequently, optimizing the candidate set in isolation and then superimposing it onto the base network can be suboptimal, since both the objective landscape and the cost-weight trade-offs are reshaped by the base Laplacian.

To address this joint selection-and-sizing problem under the spectral robustness objective, we adopt a standard SDP relaxation [5], eliminating explicit Laplacian inverses while preserving convexity. Here, this standard machinery is *repurposed* as a *certificate-producing surrogate* for instance-wise quality assessment. It is then recast as a standard cone program and solved efficiently with a first-order operator-splitting method (SCS) [15]. A certified lower bound on robustness is obtained from the relaxation, while a rounding step yields feasible discrete designs together with an a posteriori relative optimality gap. A cost-aware greedy baseline is also proposed, using rank-one updates for fast marginal-gain evaluation under the budget.

Mathematical Notations: Scalars are denoted by lowercase letters, vectors by bold lowercase letters, and matrices by bold uppercase letters. The inequality $\mathbf{A} \succeq \mathbf{B}$ means matrix $\mathbf{A} - \mathbf{B}$ is positive semidefinite; $\mathbb{S}_+^n$ is the cone of $n \times n$ symmetric

This work is supported in part by the MoST (MobilitetsLab Stor-Trondheim) project, and the PERSEUS project, a European Union’s Horizon 2020 research and innovation program under the Marie Skłodowska-Curie grant agreement No 101034240, and the Research Council of Norway under the project “Resource-aware IoT with Enhanced Intelligence and Security.”

positive semidefinite matrices. $\text{Tr}(\cdot)$ denotes the matrix trace, $\text{val}(\cdot)$ is the optimal objective value, and the superscript $*$ marks optimal variables.

II. BUDGETED GRAPH AUGMENTATION

We consider an undirected base graph $\mathcal{G}(V, E_b, \mathbf{W}_b)$ with vertex set $V = \{v_1, \dots, v_n\}$, base edges $E_b = \{e_1, \dots, e_m\}$ (with $e = \{v_i, v_j\} = \{v_j, v_i\}$), and a symmetric weighted adjacency matrix $\mathbf{W}_b$ encoding the strengths of existing links. This matrix is fixed throughout. Let $\mathbf{D}_b$ be the associated degree matrix, and define the base Laplacian as $\mathbf{L}_b = \mathbf{D}_b - \mathbf{W}_b$.

To enhance robustness, we augment the graph by selecting and sizing a subset of p candidate edges $E_c = \{e_{m+1}, \dots, e_{m+p}\}$ disjoint from E_b . Each candidate $e = \{v_i, v_j\}$ has an associated incidence vector $\mathbf{a}_e \in \mathbb{R}^n$, where $[\mathbf{a}_e]_i = 1$, $[\mathbf{a}_e]_j = -1$, and all other entries are zero. Each edge is assigned a nonnegative design weight $w_e \geq 0$, representing its strength (e.g., conductance, bandwidth, or inverse delay). To model resource constraints, each edge also carries a per-unit deployment cost $c_e > 0$, forming a cost vector $\mathbf{c} \in \mathbb{R}^p$, reflecting length, terrain, or installation overhead.

Define the augmented Laplacian

$$\mathbf{L}_a(\mathbf{w}) = \sum_{e \in E_c} w_e \mathbf{a}_e \mathbf{a}_e^\top, \quad \mathbf{L}(\mathbf{w}) = \mathbf{L}_b + \mathbf{L}_a(\mathbf{w}). \quad (1)$$

We assume the base graph is connected; hence, $\mathbf{L}(\mathbf{w}) + \mathbf{1}\mathbf{1}^\top/n \succ 0$ for all feasible $\mathbf{w}$ [16]. Lastly, the total effective resistance R_{tot} of the graph is given by [5] as follows:

$$R_{\text{tot}}(\mathbf{w}) = n \text{Tr} \left(\mathbf{L}(\mathbf{w}) + \frac{\mathbf{1}\mathbf{1}^\top}{n} \right)^{-1} - n. \quad (2)$$

To jointly select which k candidate edges to add and to assign their weights, we introduce binary variables $\mathbf{z} \in \{0, 1\}^p$ (selection) and continuous weights $\mathbf{w} \in \mathbb{R}_{\geq 0}^p$ (sizing). This yields the following mixed-integer formulation:

$$\begin{aligned} & \underset{\mathbf{w}, \mathbf{z}}{\text{minimize}} && R_{\text{tot}}(\mathbf{w}) \\ & \text{subject to} && \mathbf{c}^\top \mathbf{w} \leq C, \\ & && z_i w_{\min} \leq w_i \leq z_i w_{\max}, \quad i = 1, \dots, p, \\ & && \mathbf{1}^\top \mathbf{z} = k, \quad z_i \in \{0, 1\} \end{aligned} \quad (\mathcal{P}_0)$$

with optimal value denoted by $(R_{\text{tot}}^{\text{MI}})$.

In $(\mathcal{P}_0)$, the budget constraint $\mathbf{c}^\top \mathbf{w} \leq C$ uses per-unit weight costs $\mathbf{c}$, coupling selection and sizing through cost. The perspective box bounds $z_i w_{\min} \leq w_i \leq z_i w_{\max}$ tie weight feasibility to selection: if $z_i = 0$ then $w_i = 0$, while $z_i = 1$ enforces $w_i \in [w_{\min}, w_{\max}]$. The cardinality constraint $\mathbf{1}^\top \mathbf{z} = k$ adds exactly k links. We use $\mathbf{L}_a(\mathbf{w})$ as defined in (1); the selection vector $\mathbf{z}$ acts indirectly by gating each w_i through the perspective bounds, so an explicit $\mathbf{L}_a(\mathbf{w}, \mathbf{z})$ is unnecessary.

The combinatorial edge-selection in $(\mathcal{P}_0)$ renders the problem NP-hard [8]. To obtain a tractable surrogate that relaxes the integrality constraint and provides an instance-wise certificate, we relax the binary variables from $\mathbf{z} \in \{0, 1\}^p$ to $\mathbf{z} \in [0, 1]^p$. Nonetheless, directly evaluating R_{tot} in this

relaxed form entails factorizations/solves with the shifted Laplacian

$$\mathbf{M}(\mathbf{w}) := \mathbf{L}(\mathbf{w}) + \frac{\mathbf{1}\mathbf{1}^\top}{n} \succ 0, \quad (3)$$

which has an $\mathcal{O}(n^3)$ worst-case cost on large graphs [7]; moreover, naïve sparsification can distort the Laplacian spectrum and effective resistances [17]. To avoid any explicit (pseudo)inverse while preserving convexity, we apply a standard Schur-complement lift following [5]. Introducing an auxiliary matrix variable $\mathbf{R} \succeq 0$ yields the following SDP:

$$\begin{aligned} & \underset{\mathbf{R}, \mathbf{w}, \mathbf{z}}{\text{minimize}} && n \text{Tr}(\mathbf{R}) - n \\ & \text{subject to} && \mathbf{c}^\top \mathbf{w} \leq C, \\ & && z_i w_{\min} \leq w_i \leq z_i w_{\max}, \quad i = 1, \dots, p, \\ & && \mathbf{1}^\top \mathbf{z} = k, \quad 0 \leq z_i \leq 1, \quad i = 1, \dots, p, \\ & && \begin{pmatrix} \mathbf{M}(\mathbf{w}) & \mathbf{I} \\ \mathbf{I} & \mathbf{R} \end{pmatrix} \succeq 0, \quad \mathbf{R} \in \mathbb{S}_+^n \end{aligned} \quad (\mathcal{P}_1)$$

with optimal value denoted by $(R_{\text{tot}}^{\text{SDP}})$.

Proposition 1. Assume the base graph is connected, so that $\mathbf{M}(\mathbf{w}) \succ 0$ for all feasible $\mathbf{w}$. Then $(\mathcal{P}_1)$ satisfies $R_{\text{tot}}^{\text{SDP}} \leq R_{\text{tot}}^{\text{MI}}$. Moreover, at any optimum $(\mathbf{R}^*, \mathbf{w}^*, \mathbf{z}^*)$ of $(\mathcal{P}_1)$, $\mathbf{R}^* = \mathbf{M}^{-1}(\mathbf{w}^*)$. Consequently, $(\mathcal{P}_1)$ provides a valid lower bound; paired with any feasible discrete augmentation, it yields an instance-wise optimality gap.

Proof (sketch). Relaxing $\mathbf{z} \in \{0, 1\}^p$ to $[0, 1]^p$ enlarges the feasible set; therefore, the optimum cannot increase. The linear matrix inequality (LMI) $\begin{bmatrix} \mathbf{M}(\mathbf{w}) & \mathbf{I} \\ \mathbf{I} & \mathbf{R} \end{bmatrix} \succeq 0$ is equivalent to $\mathbf{R} \succeq \mathbf{M}^{-1}(\mathbf{w})$ by the Schur complement [18]. Since $\text{Tr}(\mathbf{R}) - \text{Tr}(\mathbf{M}^{-1}(\mathbf{w})) = \text{Tr}(\mathbf{R} - \mathbf{M}^{-1}(\mathbf{w})) \geq 0$, minimizing $n \text{Tr}(\mathbf{R})$ forces $\mathbf{R} = \mathbf{M}^{-1}(\mathbf{w})$, reproducing the trace-of-inverse objective as in [5]. $\square$

Since the relaxed SDP does not yield an integer solution, we construct a feasible design for $(\mathcal{P}_0)$ by a *rounding-and-repair* procedure. Starting from the SDP solution $(\mathbf{w}^*, \mathbf{z}^*)$ of $(\mathcal{P}_1)$, select the k largest entries of $\mathbf{z}^*$ (breaking ties by smaller unit costs c_i) to obtain $\hat{\mathbf{z}} \in \{0, 1\}^p$ with $\mathbf{1}^\top \hat{\mathbf{z}} = k$. Project the weights onto the box induced by $\hat{\mathbf{z}}$ via $\tilde{w}_i = \min\{\max(w_i^*, w_{\min} \hat{z}_i), w_{\max} \hat{z}_i\}$. If $\mathbf{c}^\top \tilde{\mathbf{w}} \leq C$, set $\hat{\mathbf{w}} = \tilde{\mathbf{w}}$; otherwise scale along the segment from $\tilde{\mathbf{w}}$ toward $w_{\min} \hat{\mathbf{z}}$ to enforce $\mathbf{c}^\top \hat{\mathbf{w}} \leq C$. If $\mathbf{c}^\top (w_{\min} \hat{\mathbf{z}}) > C$, then no feasible weights exist for this selection; one must reduce k , increase C , or modify $\hat{\mathbf{z}}$ (e.g., swap out the highest-cost edges). The pair $(\hat{\mathbf{w}}, \hat{\mathbf{z}})$ is then feasible for $(\mathcal{P}_0)$ and is used in the optimality gap below.

Remark 1. Given a feasible rounded pair $(\hat{\mathbf{w}}, \hat{\mathbf{z}})$, we have:

$$R_{\text{tot}}^{\text{SDP}} = \text{val}(\mathcal{P}_1) \leq \text{val}(\mathcal{P}_0) = R_{\text{tot}}^{\text{MI}} \leq R_{\text{tot}}(\hat{\mathbf{w}}) =: R_{\text{tot}}^{\text{RND}}.$$

We can then report the relative gap ratio $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}})/R_{\text{tot}}^{\text{SDP}}$ as a measure of solution quality. If the rounded pair is not feasible, it is not an upper bound; we first repair it to feasibility before evaluating R_{tot} .

Next, we reformulate $(\mathcal{P}_1)$ as a standard cone program and solve it using an operator-splitting method.

III. CONE-PROGRAM REFORMULATION AND OPERATOR SPLITTING

To solve problem $(\mathcal{P}_1)$, we employ conic optimization via an operator splitting method [15]. This approach handles large-scale problems with lower per-iteration cost, at the expense of more iterations to reach high accuracy. Starting from the SDP relaxation $(\mathcal{P}_1)$, we convert it into an equivalent standard cone problem by applying the scaled vectorization $\text{svec}(\cdot)$ to the symmetric variables (off-diagonals scaled by $\sqrt{2}$ to preserve trace inner products), and embed the result in the standard cone-program form:

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} \quad \mathbf{v}^\top \mathbf{x} \\ & \text{subject to} \quad \mathbf{A}\mathbf{x} + \mathbf{s} = \mathbf{b}, \quad (\mathbf{x}, \mathbf{s}) \in \mathbb{R}^{\tilde{n}} \times \mathcal{K} \end{aligned} \quad (\mathcal{P}_2)$$

where $\tilde{n} = \frac{n(n+1)}{2} + 2p$, $\mathbf{x} = [\text{svec}(\mathbf{R})^\top \mathbf{w}^\top \mathbf{z}^\top]^\top$, and $\mathbf{v} = [n \text{svec}(\mathbf{I})^\top \mathbf{0}_p^\top \mathbf{0}_p^\top]^\top$, so that $\mathbf{v}^\top \mathbf{x} = n \text{Tr}(\mathbf{R})$. The data matrix $\mathbf{A}$ and corresponding vector $\mathbf{b}$ are obtained by stacking the affine constraints of $(\mathcal{P}_1)$ (after $\text{svec}(\cdot)$ where applicable) and introducing slack variables for inequalities.

For the PSD constraint, we represent the block LMI through an affine map plus a PSD slack $\mathbf{S}_{\text{SDP}} \in \mathbb{S}_+^{2n}$, and use $\mathbf{s}_{\text{SDP}} := \text{svec}(\mathbf{S}_{\text{SDP}})$ in the standard cone form. For the inequality budget constraint upper bounded by C , we introduce a nonnegative slack variable. Similarly, for bounded edge weights and relaxed selection variables, we split the inequalities, introduce nonnegative slack variables, and obtain the corresponding blocks of $\mathbf{A}$ and entries of $\mathbf{b}$. The equality constraint on edge selection is represented using a zero-cone (i.e., slack constrained to $\{0\}$).

Finally, we have $\mathbf{A}^\top = [\mathbf{A}_{\text{SDP}}^\top \mathbf{A}_{\text{cost}}^\top \mathbf{A}_{\text{box}}^\top \mathbf{A}_{\text{card}}^\top]^\top$, $\mathbf{b} = [\mathbf{b}_{\text{SDP}}^\top C \mathbf{b}_{\text{box}}^\top k]^\top$, with $\mathbf{s} = [\mathbf{s}_{\text{SDP}}^\top \mathbf{s}_{\text{cost}}^\top \mathbf{s}_{\text{box}}^\top \mathbf{s}_{\text{card}}^\top]^\top$ representing the slack variables lying in the primal cone $\mathcal{K}$. The product cone is $\mathcal{K} = \mathbb{S}_+^{2n} \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+ \times \{0\}$, i.e., a Cartesian product of one semidefinite cone, nonnegative orthants, and a zero cone that together encode our LMI, cost, box, and cardinality constraints.

For the dual problem of $(\mathcal{P}_2)$, the dual residual $\mathbf{r} \in \mathbb{R}^{\tilde{n}}$ is constrained by $\mathbf{r} = \mathbf{0}$ and the dual variable satisfies $\mathbf{y} \in \mathcal{K}^*$ (dual cone). Employing Eq. (6) in [15] yields a single feasibility system that embeds the primal-dual Karush-Kuhn-Tucker (KKT) conditions with $(\mathbf{x}, \mathbf{s}, \mathbf{r}, \mathbf{y}) \in \mathbb{R}^{\tilde{n}} \times \mathcal{K} \times \{0\}^{\tilde{n}} \times \mathcal{K}^*$; any feasible point recovers a primal-dual optimal solution of the standard cone program. To robustly handle infeasibility/unboundedness and obtain certificates, we use the homogeneous self-dual embedding [15], [19]. Applying the ADMM-style operator-splitting algorithm in Eq. (17) of [15] to the resulting embedding yields a solution (or certificate) for $(\mathcal{P}_2)$ up to tolerance, and, hence, for $(\mathcal{P}_1)$. The method solves the optimization problem using a cached $\tilde{\mathbf{L}}\mathbf{D}\tilde{\mathbf{L}}^\top$ factorization for the quasi-definite linear system in Eq. (17) of [15]. The one-time factorization cost depends on the fill-in induced by the sparsity pattern (here determined by the self-dual embedding matrix) and the elimination ordering, and is cubic-time in the worst case for dense fill. Thereafter, the

affine projection (linear-solve) step is performed via forward-backward substitutions with the cached factors, with per-iteration cost linear in the number of nonzeros of the factors. The remaining dominant operation is the projection onto the product cone: it decomposes across cone blocks, and requires an eigenvalue decomposition for the PSD block (followed by thresholding of negative eigenvalues) with cubic cost in the PSD-block dimension, and inexpensive entrywise clipping for the nonnegative orthant with cost $\mathcal{O}(p)$ [20].

IV. GREEDY ALGORITHM

To obtain a fast feasible baseline, we employ a greedy augmentation strategy in the spirit of [9]. Rather than optimizing all edges jointly, the method adds edges sequentially by selecting at each step the candidate with the largest predicted marginal decrease in R_{tot} .

Let $\mathbf{M}$ denote the current shifted Laplacian $\mathbf{M}(\mathbf{w})$ as defined in Eq. (3) for the current augmented graph with edge-weight vector $\mathbf{w}$. Adding an edge $e = (v_i, v_j)$ with incidence $\mathbf{a} := \mathbf{e}_{v_i} - \mathbf{e}_{v_j}$ and weight $w_{(v_i, v_j)} \geq 0$ yields $\mathbf{M}' = \mathbf{M}(\mathbf{w}) + w_{(v_i, v_j)} \mathbf{a}\mathbf{a}^\top$. By the Sherman-Morrison (rank-one update) identity [21],

$$\mathbf{M}'^{-1} = \mathbf{M}^{-1} - \frac{w_{(v_i, v_j)}}{1 + w_{(v_i, v_j)} \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a}} (\mathbf{M}^{-1} \mathbf{a})(\mathbf{M}^{-1} \mathbf{a})^\top. \quad (4)$$

Define $r(v_i, v_j) := \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a}$ and $b^2(v_i, v_j) := \|\mathbf{M}^{-1} \mathbf{a}\|_2^2$. Then the marginal gain admits the closed form

$$\Delta R_{\text{tot}} = \frac{n w_{(v_i, v_j)} b^2(v_i, v_j)}{1 + w_{(v_i, v_j)} r(v_i, v_j)}.$$

To assign a trial weight under the remaining budget C_{rem} , and remaining link additions k_{rem} (with candidate edge set $E_{\text{rem}} \subseteq E_c$), we use a simple cost-aware provisional spend $s := C_{\text{rem}}/k_{\text{rem}}$ and set $w_{(v_i, v_j)} := \max \left\{ w_{\min}, \min \left\{ w_{\max}, \frac{s}{c_{(v_i, v_j)}} \right\} \right\}$. The greedy algorithm starts by computing the inverse of the current shifted Laplacian $\mathbf{M}_0^{-1}(\mathbf{w})$, at cost $\mathcal{O}(n^3)$. Then, at each step the greedy method evaluates all remaining candidates using ΔR_{tot} under the provisional weight $\{w_{(v_i, v_j)}\}$, selects the best edge, updates $\mathbf{M}_k^{-1}$ using Eq. (4), removes the chosen edge from E_{rem} , and updates $C_{\text{rem}} \leftarrow C_{\text{rem}} - c_{(v_i, v_j)} w_{(v_i, v_j)}$. The procedure repeats until k edges are added or no feasible candidate remains (i.e., $C_{\text{rem}} < c_{(v_i, v_j)} w_{\min}$ for all remaining edges), with each step costing $\mathcal{O}(pn + n^2)$. The algorithm executes with total complexity of $\mathcal{O}(n^3) + \mathcal{O}(k(pn + n^2))$. The greedy is a heuristic for $(\mathcal{P}_0)$; each marginal gain is computed exactly via Sherman-Morrison, and its objective value is compared empirically against the lower bound provided by $(\mathcal{P}_1)$.

V. SIMULATION RESULTS

To evaluate the proposed methods, we report experiments on synthetic spatial graphs and a real road network. Throughout, edge weights model *link strength or capacity*, while costs model the expense of constructing or upgrading candidate links. We report the relative gap between the $(\mathcal{P}_1)$ lower bound

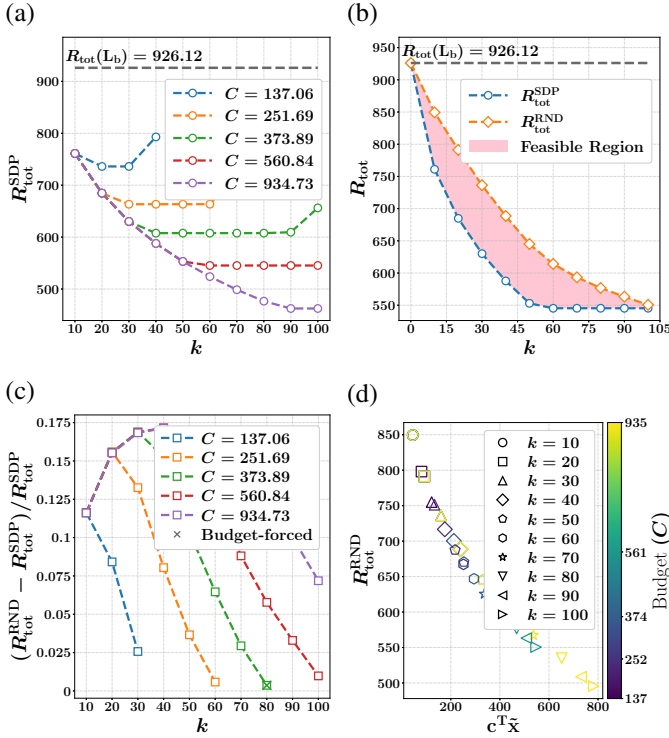


Fig. 1: (a) $R_{\text{tot}}^{\text{SDP}}$ versus k . (b) Feasible region. (c) Relative gap $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}}) / R_{\text{tot}}^{\text{SDP}}$ versus k (Remark 1) for strategic C . (d) Pareto front: $R_{\text{tot}}^{\text{RND}}$ versus spent budget for fixed k , across various cap budgets.

and feasible discrete designs as the principal quality measure; existing greedy methods [9], [10] operate under cardinality-only or unweighted regimes that require non-trivial adaptation to the cost-aware setting.

For the synthetic setting, we minimize R_{tot} via $(\mathcal{P}_1)$ on an undirected weighted base graph with $n = 100$ vertices and fixed planar coordinates. Base edge weights are drawn as $w_b = Z$, where $\ln Z \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$. Thus w_b follows a lognormal distribution, modeling broadly skewed edge weight statistics with heterogeneous link capacities in spatial networks, and commonly observed in weighted graphs [22]. Added links are sampled from the upper tail (or scaled-up samples) to represent engineered upgrades with higher weights. We set bounds on candidate-edge weights as $w_{\min} = Q_{0.75}$ and $w_{\max} = Q_{0.95}$, where Q_p denotes the p -quantile of the lognormal distribution used to draw the base edge weights. To obtain a tractable and plausible candidate set, we partition the planar vertex coordinates into K regions using k -means clustering and obtain E_c as the union of (i) intra-region k_{in} nearest-neighbor links and (ii) a sparse inter-region backbone, thereby capturing the common *hub-and-spoke* structure of practical network design. We set the clustering parameters to $K = 3$ and $k_{\text{in}} = 3$.

Candidate-edge costs follow a fixed-charge network design model [23]: $\mathbf{c} = c_{\text{fixed}} \mathbf{1} + \beta \mathbf{d}_c$, where c_{fixed} is a fixed per-link overhead (endpoints/installation) incurred when a link is

selected and β scales the Euclidean length-dependent penalty. The entries of $\mathbf{d}_c$ are normalized lengths $d_e / \bar{d}$, where d_e is the candidate-edge Euclidean length and $\bar{d}$ its mean across candidates, ensuring compatibility across instances and consistent units. We set $c_{\text{fixed}} = 1.5$ and $\beta = 0.6$.

Rather than specifying budget values a priori, we strategically deduce the budgets from the cost structure of the candidate edge set. Given the set of candidate edges and its cost model, the minimum feasible budget to add k edges is $C_{\min}(k) := w_{\min} \sum_{i=1}^k c_{(i)}$, where the sum is taken over the k cheapest candidates. We select five budget levels centered on the anchor $C_{\min}(k_{\max})$, which is the smallest budget permitting all $k_{\max}$ edge additions. Two budgets below the anchor are set at the feasibility thresholds of intermediate k -values (approximately $\frac{1}{3}$ and $\frac{2}{3}$ of $k_{\max}$), creating regimes where the budget constraint is binding for large k but slack for small k . Two budgets above the anchor (at $1.5\times$ and $2.5\times$) probe the budget-relaxed regime where the constraint becomes progressively less restrictive. This strategy ensures that the experimental budget spans from tightly constrained to unconstrained optimization, regardless of the graph geometry or cost parameters. The resulting budget set is $C = \{137.06, 251.69, 373.89, 560.84, 934.73\}$.

Using these budgets, Fig. 1a plots $R_{\text{tot}}^{\text{SDP}}$ against k , and Fig. 1b shows the feasible region at budget cap $C = 560.84$. Based on Proposition 1, Fig. 1c reports the relative gap ratio for augmentation, and in Fig. 1d the cost and effective resistance trade-off across all (k, C) combinations is visualized via the Pareto front. Figs. 2b, 2c, and 2d compare R_{tot} versus k for three of the strategic budgets. In all cases, the SDP relaxation provides a lower bound that decreases rapidly for small k before saturating as the marginal gain per edge diminishes. As the budget tightens, we observe that: (i) the SDP relaxation is feasible for all k at larger budgets, but becomes infeasible from $k = 80$ onward at $C = 251.69$; and (ii) the greedy algorithm exhausts its budget early on after only 61 edges at $C = 251.69$, compared to $C = 373.89$ and the full $k_{\max} = 100$ at $C = 560.84$. At the anchor budget $C = 373.89$, which corresponds to the minimum cost of installing $k_{\max} = 100$ edges at weight $w_{\min}$, all methods remain feasible but operate near the boundary: the relaxation curve flattens and begins to rise for large k because the budget forces many added edges toward $w_{\min}$, reducing marginal benefit. In Fig. 2b the shaded red region and bold cross markers identify k -values where the budget is insufficient to feasibly add k edges at minimum weight, causing the SDP solver to declare infeasibility and return no improving solution ($R_{\text{tot}} > R_{\text{tot}}(\text{L}_b)$). These results confirm that the strategically selected budget levels effectively capture the three regimes of interest: budget-relaxed, marginally feasible, and budget-constrained.

To assess performance on real-world data, we evaluate $(\mathcal{P}_1)$ on the California road network [24], with $R_{\text{tot}}^{\text{SDP}}$ vs. k reported in Fig. 2a. The dataset provides an undirected graph, and we set all base edge weights to one. We form the base instance by randomly sampling $n = 500$ nodes and extracting the largest

connected component. To construct the candidate set E_C we consider all possible node pairs in the graph and exclude those already present in the dataset, retaining only the unassigned edges for nodes in the base graph. The candidate edge weights are bounded in $[1, 5]$, and we sweep β at budget $C = 150$.

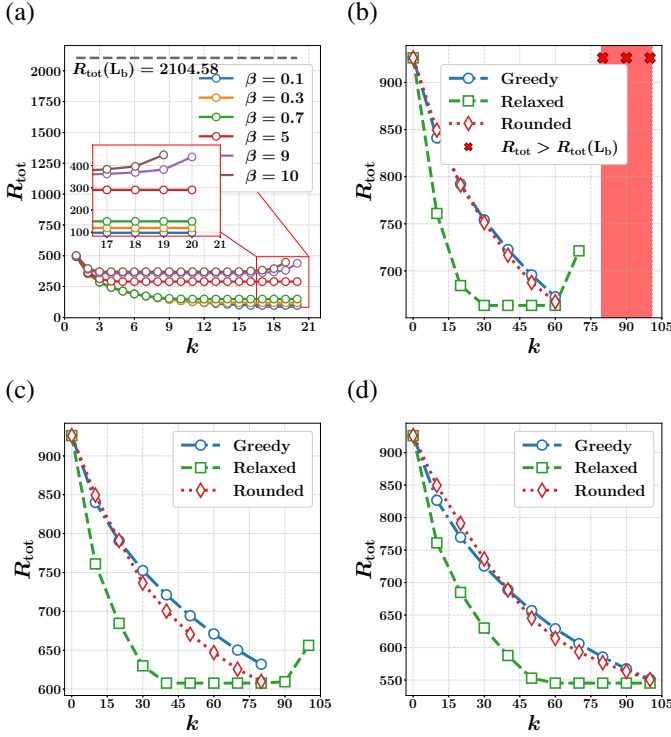


Fig. 2: **(a)** Real-world California road network; synthetic random spatial network at three strategic budgets: **(b)** $C = 251.69$, **(c)** $C = 373.89$, **(d)** $C = 560.84$.

Fig. 2a shows that $R_{\text{tot}}^{\text{SDP}}$ decreases monotonically as edges are added, and eventually the curve flattens, indicating diminishing returns. Increasing β lifts the curve, and for $\beta \in \{9, 10\}$, the curve starts bending upward. This indicates that the solution drives many candidate weights toward $w_{\min}$ and increases $R_{\text{tot}}^{\text{SDP}}$. Lastly, for $\beta = 10$, adding 20 edges renders $(\mathcal{P}_1)$ infeasible.

VI. CONCLUSION

This paper presented an optimization framework for enhancing the robustness of an existing network through budget-constrained edge augmentation. It elucidates the interplay of the cap budget and effective resistance for augmenting k edges into a base graph. The relative gap ratio between the relaxed and rounded solutions quantified the optimality gap and the quality loss induced by discretization. To improve computational efficiency, we also introduced a greedy baseline that exploits rank-one updates, trading solution quality for computational gain relative to solving the relaxation. Importantly, the relaxed semidefinite formulation is shown to lower-bound any feasible discrete augmentation. The Pareto front characterizes the inherent trade-off between effective

resistance and budget expenditure for fixed edge additions. Finally, experiments on a random spatial network and a real-world road network dataset validated the effectiveness and scalability of the framework.

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